Determination of dosimetric quantities of ^{137}Cs, half lifetime of ^{137m}Ba, test for Poisson-distributed radioactive decay

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The dose rate of a ^{137}Cs-compound was determined (i) directly above its container to be $(6,32 \pm 0,86)\,\mu$Sv/h and (ii) at a distance of 1 m to be $(0,22 \pm 0,13)\,\mu$Sv/h. The isotope was hosted in an aluminium chamber in such a way that only γ-quanta could escape. Furthermore, a Geiger-Müller tube was investigated. The optimal operating voltage was found to be $(520,0 \pm 4,0)$ V. By conducting a χ^2-test it was concluded that for both gate times (1 s and 10 s) of the GM-counter, the registered emission events caused from an uranium oxide coated tile are Poisson-distributed with a 5 % error probability. Also it was shown that the distribution yielded for a gate time of 10 s can be approximated by a Gaussian distribution. Finally, the half-lifetime of ^{137m}Ba was determined to be $(153,0 \pm 3,1)$ s.

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1 Introduction

The study of radioactivity is an essential part of the training for anyone aspiring to be a physicist for a number of reasons. First, it is a fundamental characteristic of a large part of nuclei like uranium and radium, that they stochastically decay by their specific mean life-time. Second, a rich spectrum of applications emerges from radioactive isotopes, e.g. the ^{14}C-method that is commonly used today to determine the date of death of organic matter. Third, the ionizing properties of radioactive radiation poses a risk to both the health of humans and their living environment. The amount of damage done by such rays is called equivalent dose and is measured in the unit *Sievert*, 1 Sv=1 J/kg which relates the energy that is deposited in the medium (scaled with some rating factor depending on the kind of radiation) to the medium's mass.
In our experiment we used a dosimeter to quantify the equivalent dose emitted by ^{137}Cs per time unit. Since this isotope of caesium also decays into the meta-stable γ-radiator ^{137m}Ba, we deployed a Geiger-Müller counter to detect radiated γ-quanta which allowed us to calculate the half-life of this barium nucleus. With the same apparatus we finally examined the probability distribution of registered radiation events of a tile covered with a film of uranium oxide by running a χ^2-test, checking whether the resulting histogram can be described by a Poisson- or Gaussian distribution.

2 Background radiation and radiation exposure

Since there always is a certain amount of radiation even if no radioactive material is directly exposed to our measuring devices, we have to consider what is called "background radiation". The main sources of this kind of radiation are for example terrestrial sources like potassium, uranium or airborne radon. Besides, cosmic rays, typically consisting of positively charged ions or protons from outside of our solar system contribute to it. To estimate the radiation exposure during the time spent in the laboratory we placed the dosimeter (`RADEX RD 1530`) in a representative spot and set it to long time measurement duration. In 15-minute intervals we logged the measured dose rate until our tasks were done. We determined the mean value of our collected dose rates to be $(0,103 \pm 0,011)\,\mu$Sv/h which, estimating our stay to be of three hours, results in a radiation exposure of $(0,308 \pm 0,037)\,\mu$Sv. The standard errors are derived from our measurement series. Assuming this value to

be constant during one year of staying at the place of our practical training, the entire dose of radiation would be $(0,898 \pm 0,092)$ mSv. Even under this conditions, one would not surpass the annual radiation limit of 1 mSv given by the legislation [3].

3 Dosimetric measures

3.1 The radioactive isotope ^{137}Cs

Caesium-137 is a radioactive nucleus and thus decays statistically. Radioactive rays can be classified into three different kinds: α-, β- and γ-radiation. The former consists of $^4_2He^{2+}$ ions and due to their comparatively large size they can be easily shielded off by any matter; hence they only travel a few centimeters in air, but have strong ionizing properties. The midst has two varieties, β^- and β^+: The first consists of electrons, the second of its antiparticle, the positron. Depending on its energy, β-radiation can travel about a meter in air. It is less ionizing than α-particles. Finally γ-rays represent the lower end of the wavelength spectrum of electromagnetic radiation. They are best absorbed by high-Z (high atomic number) materials like lead but one still requires multiple centimeters of eligible shielding material to be fully protected.
The prominent decay channel of Caesium-137 is via β-decay into the meta-stable isotope Barium-137m, that relaxes under the emission of a γ into the stable isotope Barium-137. A direct β-decay into the same isotope is also possible, but only with a branching ratio of 5 %.
In the first part of our experiment we analyzed the equivalent dose output per unit time of a Cs-137 compound prepared in a stick of aluminium, while its walls were thick enough to shield off the emitted β-radiation.

3.2 Dose rate of ^{137}Cs

We used the `NEVA` dosimeter at 5 s measuring interval to examine the dose rate of γ-radiation emitted by the ^{137}Cs preparation. Placing the apparatus above the gap of our lead cube, we determined the dose power of the collimated γ-beam to be $(6,32 \pm 0,86)\,\mu$Sv/h which would result in an equivalent dose of $(25,3 \pm 3,4)\,\mu$Sv during four hours. This corresponds to $(0,505 \pm 0,069)$ % of the annual limit of 5 mSv. Therefore at such close distance to our radioactive material, one could stay for a total of $(33,0 \pm 4,7)$ d to absorb this critical amount of radiation.

To examine the influence of the distance kept to the radiator on the absorbed dose we placed the lead cube on the floor and the dosimeter on the table with its sensitive window facing the radio source, roughly one meter away from the unshielded ^{137}Cs-sample and measured the dose rate to be $(0,22 \pm 0,13)\,\mu$Sv/h. Within four hours, this corresponds to an equivalent dose of $(0,87 \pm 0,50)\,\mu$Sv. In this case we neglected the background radiation, since it is covered within the margin of error of the computed dose rate.

3.3 Activity of ^{137}Cs

The decay of radioactive materials is described by an exponential law that can be derived by considering a constant probability of decay for every single nucleus. The number of radioactive nuclei $N(t)$ decreases over time and if the number of unstable atoms inside a material is very high, it abides by the rate equation

$$\frac{dN(t)}{dt} = -\lambda N(t), \tag{1}$$

where λ is the decay constant, which takes into account the constant probability of a single-nucleus decay. Solving this first-order differential equation yields the radioactive law

$$N(t) = N_0 e^{-\lambda t} = N_0 e^{-t/\tau}. \tag{2}$$

Here N_0 is the initial amount of radioactive material and $\tau = 1/\lambda$ is its mean lifetime. To get a time-dependent expression for the activity $A(t)$ of decaying matter, the condition $A(t) = -\frac{dN(t)}{dt}$ outputs the activity law

$$A(t) = A_0 e^{-\lambda t}, \tag{3}$$

where $A_0 = \lambda N_0$ is the initial activity of our radioactive material. It can be seen that after some time only half of the initial radioactive nuclei are left. This specific time is called half lifetime T_H. Solving the condition $N(T_\mathrm{H}) = \frac{1}{2}N_0$ we get:

$$T_\mathrm{H} = \frac{ln(2)}{\lambda} = \tau\, ln(2) \tag{4}$$

According to its manufacturer, the date of production was *August 1978* and its activity at that time was $3,70 \times 10^6$ Bq. Since its half-life time equates $30,2$ a, the activity as of today follows with $A_\mathrm{today} = 1,36 \times 10^6$ Bq.

4 Geiger-Müller counter characteristics

In this section we evaluated the counting characteristic of the Geiger-Müller tube connected to a `LEYBOLD Digital` counter. We deployed both for all of our measurements in this paper. The detection window of the tube was put directly above the ^{137}Cs preparation to catch as much events as possible. This helps us to determine the critical voltage from which on our

tube started to detect these events. For this part no shielding was used. Now, beginning with a voltage of 500 V we set the gate time to 10 s and started the measurement. We then increased the operating voltage in constant steps of 8 V until we reached 600 V. The uncertainty of the voltage is estimated to be $4V$ since the digital display's smallest unit of scale was 4 V.

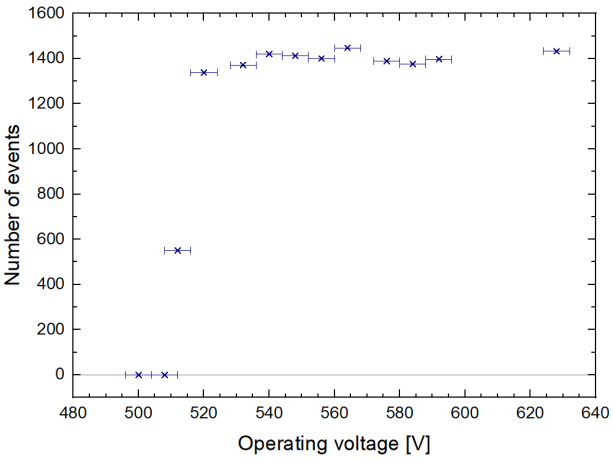

Figure 1: Number of events counted within 10 s gate time directly above a ^{137}Cs-sample, plotted depending on the operating voltage of a Geiger-Müller tube.

In fig. 1 the resulting diagram is shown. We noticed, that at a critical operating voltage of 512 V our tube started detecting some events, but did not reach the full sensitivity yet. At 520 V operating voltage it is valid to assume the Geiger-Müller counter is at its plateau of sensitivity. Above that threshold increasing the voltage does not affect the number of events detected. It has to be added though that there exists a critical voltage at which the continuous discharge of the filling gas of the tube onsets, resulting in permanent damaging of the detector.

5 Null effect

As discussed in chapter 2, we need to take into account the background radiation that influences every measurement of decay events. Now we evaluated this phenomenon more precisely by using the same Geiger-Müller counter. We placed the counting tube in the middle of our workplace as far away as possible from radiating material. Then we started our measurement with a gate time of 10 s and recorded the events for a total of five minutes. We computed the mean number of events to be (37 ± 13) 1/min. The error was set to be the standard error from our series of measurements.
Finally, we performed the same measurement with the counting tube placed inside the lead shielding to determine the amount of background radiation absorbed by the shielding. For this setup we found a rate of $(16,0 \pm 9,0)$ 1/min. We thus found that the lead shield blocks off (42 ± 28) % of the incoming background radiation. It should be noted that the big relative errors are caused by the statistical distribution of the measured events. Within the setup without the lead shield, they covered a span of 2 to 11 events during 10 s gate time, which we think ought to be normal.
Theoretically, it is expected that γ radiation inside of a dense material interacts microscopically with the matter in three different ways, and as a consequence, loses its energy. The main interaction processes for γ-photons are:
1. the core-photoeffect, which is the dominant process for energies up to 1 MeV and is proportional to Z^4 of the absorbing material;
2. Compton scattering which belongs to an energy spectrum of 1 MeV to 5 MeV;
3. pair production for photon energies above 5 MeV, which is proportional to Z^2. Since these interactions increase for increasing nuclear masses, and also due to its favorable cost/use ratio, lead is regularly used to shield off γ-rays. Despite being a good absorber, it does not block off every photon, as can be interpreted from the roughly 40 %-reduction of background radiation, calculated earlier in this section. Therefore the commonly used dimension to describe the absorbing properties in relation to γ-radiation is the half-value thickness; the thickness of the material at which the intensity of radiation entering is reduced to half.

6 Half lifetime of ^{137m}Ba

To determine the half lifetime of Ba-137m, the metastable product of beta-decaying ^{137}Cs, we used an `Cs-137/Ba-137m Isotope Generator` [4] allowing us to elute Ba-137m by injecting hydrochloric acid into the compound with a syringe. This step was performed by Dr Kießling. Then we placed the solution inside a plastic fixture and adjusted the Geiger-Müller counting tube directly in front of it to capture as much events as possible. We set the gate time to 10 s and started the measurement immediately after the solution was eluted. The number of events were screened over a time of roughly 30 min. The result is shown in fig. 2.

To determine the half lifetime of Ba-137m, we fitted a curve to the data using the function

$$A(t) = y_0 + A_0 e^{-\lambda t}, \tag{5}$$

which is a way of describing the activity of some radioactive material (as discussed in section 3.3), but now with an offset of y_0 that will be discussed later on.

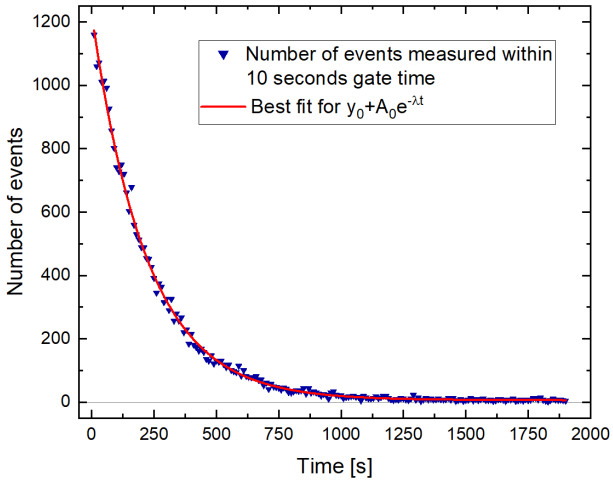

Figure 2: Number of events counted within 10 s gate time directly in front of the eluted Ba-137m; plotted against the time. An exponential fit for $y_0 + A_0\, e^{-\lambda t}$ was adjusted to our data, where y_0 represents the underground caused by background radiation, A is the activity of Ba-137m and λ its decaying constant.

We evaluated the decaying contant to be $\lambda = (4,530 \pm 0,037) \times 10^{-3}\,1/\mathrm{s}$ which, referring to eq. (4), results in a half lifetime of

$$T_{\mathrm{H}}(^{137\mathrm{m}}\mathrm{Ba}) = (153,0 \pm 1,3)\,\mathrm{s},$$

where the errors were taken from the exponential fit. It can thus be said that within its margin of error, the half lifetime aligns well with the literature value of $T_{\mathrm{H,\,lit.}} = 153,12\,\mathrm{s}$ [2]. It should be noted that we decided to ignore the null effect on the measurements for this experiment since its existence would not have affected the half-lifetime, as just an additional constant error would have featured for every data point measured.
On the contrary, with knowledge of the actual Ba-137m-half lifetime from literature, we were able to choose the gate time to be 10 s, a small fraction of the half lifetime. This information facilitated the execution of the experiment, yet in reality it is not given. Running our measurements sufficiently long, we should only detect background radiation, since nearly every Ba-137m would have transitioned into the ground state by emitting a photon. The constant y_0 in the fit-function accounted for this. We determined its value to equate $y_0 = (37,4 \pm 8,0)\,1/\mathrm{min}$, which, with an even greater precision, reproduces our previous value for the null effect that we evaluated to be $(37 \pm 13)\,1/\mathrm{min}$ in section 5. As mentioned before, in reality, not knowing the half lifetime beforehand, determining it can be a more demanding process.

7 Counting statistics

7.1 Gate time 1 s

In this part we used a χ^2-test to check whether the occurrences of number of detection events in a one-second time window is in accordance to a Poisson-distribution. For this we positioned a Geiger-Müller tube inside a chamber with lead walls of approximately 5 cm thickness, together with a roughly 15 cm × 15 cm tile coated on one side with uranium oxide. We closed the chamber by placing two bricks of lead of equal thickinesses in front of the aperture. With this setup the experiment can be assumed to be isolated from outer radioactive influences; thus, the null effect was ignored for this and the following experiment for 10 s gate time.

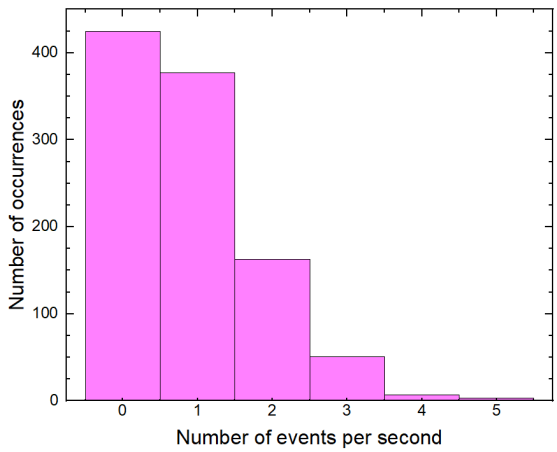

Figure 3: Histogram of number of occurrences depending on the number of detected γ-quanta by a Geiger-Müller tube with one second gate time. The radio source was an uranium oxide coated tile in a lead chamber.

To conduct a χ^2-test we made 1026 measurements with a gate time of one second and formulated the null hypothesis as H_0 : "Our data set is a sample of Poisson-distributed population with an average of $\mu = (0,88 \pm 0,93)$ events/1 s." For the error σ_μ we used the sample standard error; its large size stems from the large spread of values in our data set. Its sample variance reads $\sigma^2 = 0,866$. We chose the significance level to be $0,95$. A histogram of the acquired data is depicted in fig. 3.

The result of the test was $\chi^2 = 0,79$ (for details see Appendix, chapter 10). A contingency table requires for this situation $\chi^2_{\mathrm{max}} = 7,81$, thus we accept H_0: With a 5 % error probability the data shown in fig. 3 is Poisson-distributed.

7.2 Gate time 10 s

Proceeding analogously, we chose the same null hypothesis for the 1001 data points recorded with a gate time of ten seconds, but otherwise the same setup. The obtained data is shown as a histogram in fig. 4. This time the average was $\mu = (8,9 \pm 3,1)$ events/10 s and the variance $\sigma^2 = 9,622$, leading to $\chi^2 = 16,61$. Since $\chi^2_{\max} = 25,00$, we again accept H_0 with a $0,95$ security level.
We note that the difference between $\chi^2_{\max}$ and χ^2 is larger for the higher gate time of ten seconds. This observation is explained by the total duration of each measurement: $1065 \times 1\,\mathrm{s} < 1001 \times 10\,\mathrm{s}$, the measurement for the longer gate time took 10 times longer than the other. The more data we collect, the better random statistical fluctuations neutralize themselves, thus the more precise the result.

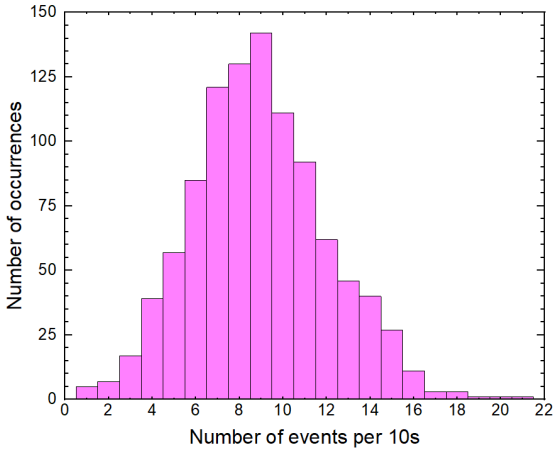

Figure 4: Histogram of number of occurrences depending on the number of detected γ-quanta by a Geiger-Müller tube with ten second gate time. The radio source was an uranium oxide coated tile in a lead chamber.

For $\mu \geq 9$ the Poisson-distribution approximately corresponds to the normal distribution. Since our average value μ in this case is ≈ 9, we obtain a quantitative statement on how well our data is described by a normal distribution by plotting the cumulative probability of detecting at most n occurrences depending on n with a probability-scaled y-axis. This has been done in fig. 5. For a perfect normal distribution, we would expect a straight line with a correlation coefficient of $r = 1$. Yet, since our μ is below the limit of 9, we do not expect a perfectly linear correlation.

In this case, the linear regression of the data shown in fig. 5 yielded $r = 0,99184$. This agrees with the qualitative fact that the fit we did appears to describe our data well. It has to be added though that the deviations from the regression are not randomly spread:

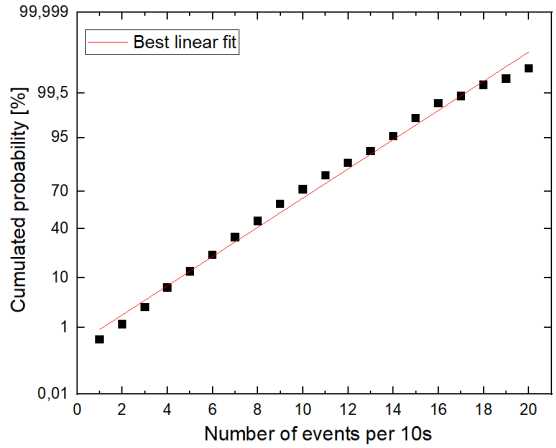

Figure 5: Cumulative probability of detecting n events with a Geiger-Müller tube with 10 s gate time. To check whether a normal distribution is present, a linear regression has been added; a perfectly normal distributed data set would equate a straight line in this diagram. The experiment was done with an uranium oxide coated tile in a lead chamber.

For low and high numbers of events per gate time the fit overestimates the data, while in between the line runs below the measured values. We conclude that for $\mu \approx 9$ as in our case, the Poisson-distribution can be approximated by a normal distribution. The greater the μ is, the smaller are the deviations.

8 Responsiveness of the Geiger-Müller counter

The responsiveness Λ of a detector is defined as the ratio of detected particles compared to the number of incident ones. It serves as a measure of efficiency/accuracy. If some detector has $\Lambda = 1$, no particle can penetrate it without being observed whereas another detector with $\Lambda = 0$ would be useless since it would not find anything!
To obtain a quantitative result for the responsiveness of the GM-counter we employed in our measurements, we started with the activity of our ^{137}Cs-compound at the time of the experiment of $A = 1,36\,\mathrm{MBq}$ (already obtained in chapter 3) to derive the total number of particles emitted by same. We then placed the GM-counter in three different distances r from the radiator and measured the number of particles detected. Since the geometry of the aluminium stick with the caesium compound forced us to use a ruler "in-air" to find r, we estimated its error to $\sigma_r = 0,2\,\mathrm{cm}$. Finally, we computed the responsiveness Λ by assuming an even distribution of the emission spectrum into all directions (spherical symmetry), given that the sensitive window had an area of $11\,\mathrm{cm}^2$. Done this way,

the denominator of Λ is equal to the current activity times the area of the detector window divided by the area of a sphere with radius r. The obtained results are listed in table 1 below.

r [cm]	ν [1000/s]	Λ [%]
$7,4 \pm 0,2$	$21,8 \pm 1,2$	$0,591 \pm 0,016$
$24,7 \pm 0,2$	$1,952 \pm 0,032$	$0,709 \pm 0,081$
$47,5 \pm 0,2$	$0,5279 \pm 0,0044$	$0,88 \pm 0,11$

Table 1: Responsiveness Λ of a Geiger-Müller counter depending on the radial distance r to a γ-emitting Cs-137 compound if ν particles penetrate it per second.

Since the γ-photons are barely shielded off by air, we anticipated a constant responsiveness regardless of the distance r. Thus, this result is unexpected. The dissonance can be resolved by taking a look into the modus operandi of the detector: [1] explains that due to its construction a discharge pulse renders the GM-tube insensitive for approximately 10^{-4} s. In this time span, sometimes referred to as "dead time", the tube is blind to the arrival of any particles. As can be seen in the first row of table 1, for the shortest source-detector distance, we received a higher number of particles than we could detect, resulting in a drop of the response function beyond the error margins. For the other two distances we did not exceed the tube's measurements limits, thus the responsivenesses agree with each other within their margins of error.
We note that we assumed the particles to incide with a constant frequency. Even for frequencies lower than the bound $\nu_{max} = 10^4$ particles/s given by the dead time, one might measure a lesser responsiveness due to the fact that the momentary frequency might be higher than ν_{max} although, on average, it is not ($\nu_{\text{avg}} < \nu_{\text{max}}$). Technically one always has to account for this, yet finding a correcting factor is experimentally hard. Finally, it has to be added that the responsiveness is less than expected: On average, less than one in a hundred particles is detected by the GM-tube.

9 Summary

A radioactive ^{137}Cs-compound was examined. With a dosimeter we measured the dose rate caused by γ-emission via the secondary decay process to be $(6,32 \pm 0,86)\,\mu$Sv/h directly above an aperture of lead cube the isotope was stored in. In 1 m distance, the dose rate decreased to only $(0,22 \pm 0,13)\,\mu$Sv/h. Consequently, one should always try to maintain maximum distance to any sources of radioactivity for her own security. To get a quantitative estimate of the total dose absorbed during the time spent in the laboratory, we placed a dosimeter set to long-time measurement at a representative spot and calculated the accumulated dose of radiation to be $(25,3 \pm 3,4)\,\mu$Sv, equal to $(0,505 \pm 0,069)\,\%$ of the annual limit given by the legislation.
The null effect was gauged with a Geiger-Müller tube while all the sources of radioactivity were removed from the vicinity. We were left with a background activity of $(37 \pm 13)\,1/\text{min}$. We were able to reduce this value by $\approx 42\,\%$ by enclosing the tube in a lead chamber with walls of 5 cm thickness. We also measured the responsiveness of the detector for γ-photons by calculating the ratio of particles detected to the incident ones, while varying its distance to the radioactive source. For the shortest distance we surpassed the limits of detecting speed of the tube, hence the result for this distance is to be neglected. Thus the responsiveness is the average of the results of the other distances we measured: $\Lambda = (0,79 \pm 0,14)\,\%$.
In the last part, we examined the distribution of detected events for a tile painted with uranium oxide on one side. By conducting a χ^2-test, we were able to show that for both gate times (1 s and 10 s), the occurrences of n events detected with the tube were Poisson-distributed within a 5 % error probability. Furthermore, we examined how justified assuming that a Poisson-distribution approaches a Gaussian distribution for average values $\mu \geq 9$ is. Although the average of our sample read $\mu = (8,9 \pm 3,1)$ events/10 s which is less than the critical value of 9, we still found good agreement between the distribution functions.

10 Appendix

I in 1/s	measured occurrences	Poisson-expected occurrences	χ^2
0	425	$427,18$	$0,01112$
1	377	$374,30$	$0,01944$
2	163	$163,99$	$0,00592$
3	51	$47,90$	$0,20120$
≥ 4	10	$12,64$	$0,55011$
Σ	1026	1026	$0,78780$

I in 1/s	measured occurrences	Poisson-expected occurrences	χ^2
$\leq$2	12	6,508853346	4,632565826
3	17	15,54773484	0,135651536
4	39	34,77660169	0,51290501
5	57	62,22961934	0,439483942
6	85	92,79528318	0,654844058
7	121	118,6063303	0,04830817
8	130	132,6471396	0,052826982
9	142	131,8667757	0,778681624
10	111	117,9819024	0,413173208
11	92	95,96275705	0,16364102
12	62	71,54865569	1,274333174
13	46	49,24227775	0,213482509
14	40	31,46951902	2,312367905
15	27	18,77063019	3,607898442
16	11	10,49636389	0,024165448
17	3	5,524207261	1,153400297
$\geq$18	6	5,025348825	0,189030642
Σ	1001	1001	16,60675979

Table 2: *Top*: $\chi^2 - Test$ to check whether the number of occurrences of measured events per 1 s gate time is Poisson distributed. The classes represented in the first column have been chosen such that the expected number of occurrences is at least equal to 5. The last row represents the total sum of the column and consequently the middle two columns correspond to the total number of data points. The bottom right cell is the total value for χ^2 that was used in chapter 3. The test considered three degrees of freedom.
Bottom: Same as above, but for 10 s gate time and 15 degrees of freedom.

References

[1] Versuchsanleitung Fortgeschrittenenpraktikum Modul C1-Versuch 46, Dosimetrie; Halbwertszeit; χ^2-Test, Würzburg, 2021

[2] Half-life time of ^{137m}Ba, National Nuclear Data Center, last visited on 20.03.2021, https://www.nndc.bnl.gov/nudat2/chartNuc.jsp

[3] Versuchsanleitung Fortgeschrittenenpraktikum Modul C1-Versuch 46, Begriffsbestimmung Strahlenschutz, Würzburg, 2021

[4] https://www.ezag.com/fileadmin/ezag/user-uploads/isotopes/isotopes/Isotrak/isotrak-pdf/Operation_manuals/Isotopengenerator_2011-08_A5.pdf, last visited on 17.03.2021